

PRISM-UNET: A PHYSICS-GUIDED, PROJECTION-REGULARIZED, INFORMED STATE-SPACE MULTIREOLUTION UNET FOR MEDICAL IMAGE TRANSLATION

Peeyush Kumar Singh, Inam Ul Haq Gulzar, Sneha Singh, Aditya Nigam

School of Computing and Electrical Engineering, Indian Institute of Technology, Mandi, India

ABSTRACT

The advent of Generative Adversarial Networks (GANs) has streamlined the synthesis of medical images across different modalities, a critical requirement in various clinical and practical settings. The cross-modality image translation aids clinical capabilities: while low-to-high dose CT generation subsides the radiation exposure, synthesizing different MRI sequences (T1w, T2w, FLAIR) from a single acquisition, reduces the scan time and cost significantly. To address this critical need, we propose PRISM-UNet, a GAN architecture built on a UNet-inspired framework with Mamba-integrated generator that leverages state-space modeling and linear scaling. Additionally, a carefully designed loss function incorporating physics-based constraints ensures anatomically faithful and modality-preserving synthesis. Extensive experimental evaluations and comprehensive image quality assessments validate the practical effectiveness and clinical potential of PRISM-UNet, underscoring its readiness for real-world deployment and its ability to significantly improve medical imaging workflows.

Index Terms— Medical image translation, GAN, State-Space models, Cross-modality, Physics-guided

1. INTRODUCTION

Advanced medical imaging techniques such as Magnetic Resonance Imaging (MRI) and Computed Tomography (CT) have significantly improved disease diagnosis by providing detailed views of internal organ structures. Medical image analysis is now central to diagnosis, treatment planning, and disease monitoring. While MRI excels at soft tissue contrast and CT provides tissue density information, both are often required for accurate diagnosis and radiation therapy planning. However, acquiring multiple modalities is costly, time-consuming, and sometimes infeasible due to patient-specific limitations thereby motivating automated cross-modality synthesis.

Recent advances have demonstrated the success of GANs in generating synthetic medical images[1]. Some approaches leverage unpaired data[2] using CycleGANs[3, 4], while others combine paired-unpaired learning and transfer learning[5]. Medical image translation faces three key challenges: (i) CNNs have limited global context due to local receptive fields, (ii) Transformers suffer from quadratic attention complexity for 3D volumes, and (iii) standard GANs often produce anatomically inconsistent outputs that violate acquisition physics.

To address these, we propose PRISM-UNet, a GAN framework with a Mamba-based UNet generator that provides: (1) linear-complexity long-range modeling via state-space layers with bidirectional scanning, (2) multiresolution guidance for fine detail capture, and (3) physics-based losses (k-space for MRI, Radon transform for CT) ensuring acquisition-faithful synthesis.

Extensive experiments across two datasets and four synthesis tasks demonstrate PRISM-UNet’s effectiveness. We compare against state-of-the-art methods including 3D-mDAUNet[6], ResViT[7], Ea-GAN[8], and ALDM[9], showing consistent improvements through efficient global context modeling while maintaining multi-scale architectural guidance.

The main contributions of this work are:

- **Novel Architecture:** We introduce **PRISM-UNet**, the first physics-guided, Mamba-based framework for medical volume translation. It integrates Mamba layers within a UNet generator to efficiently capture long-range dependencies while retaining spatial detail.
- **Physics-Guided Learning:** Unlike purely data-driven GANs, PRISM-UNet integrates imaging physics through projection domain and reconstruction losses, ensuring anatomically faithful synthesis.
- **Model Evaluation:** On multi-site, multimodal datasets, PRISM-UNet achieves state-of-the-art performance, with higher structural similarity and lower NMSE, while preserving anatomical fidelity and modality-consistent textures.

2. RELATED WORKS

Early work on medical image translation primarily relied on CNN-based encoder-decoder architectures [10, 11]. While effective, CNNs are limited by their local receptive fields and struggle to capture long-range dependencies. GANs became popular for their ability to generate realistic outputs. Variants such as MedGAN [12] and its unsupervised extensions [13] improved synthesis, but adversarial training remains unstable and prone to mode collapse, requiring careful hyperparameter tuning.

Transformers, with their global attention mechanisms, have also been explored. Some works use them as feature extractors [14, 15, 16], while hybrid CNN-Transformer models like ResViT [7] integrate them more deeply. However, their quadratic attention complexity limits scalability to large 3D medical volumes. Diffusion models have recently shown impressive results in high-fidelity image generation. Adaptations such as Med-DDPM [17] and ALDM [9] demonstrate strong performance for MRI and CT, but their iterative denoising process is computationally expensive and slows inference, making clinical deployment challenging.

State-space models (SSMs), especially Mamba, provide a promising alternative by reducing complexity to linear time. Recent studies [18, 19] have shown their effectiveness in segmentation, suggesting strong potential for image translation which is an area still dominated by CNN and Transformer-based methods [7, 8, 6].

3. METHODOLOGY

3.1. Data Preprocessing

All volumes undergo normalization via min-max scaling to $[0, 1]$. For CT, volumes are first clipped to $[-1024, 3000]$ HU and then normalized. All volumes are resampled to the dataset's average voxel spacing which ensures consistent geometric spacing. Due to GPU memory constraints, random 96^3 patches are extracted during training. At inference, overlapping patches are aggregated and combined via sliding-window inference with Gaussian-weighted blending.

3.2. Model

3.2.1. State-Space Models

State Space Models (SSMs) [20, 21] map input sequences $x(t) \in \mathbb{R}^L$ to output sequences $y(t) \in \mathbb{R}^L$ through hidden states $h(t) \in \mathbb{R}^N$. The continuous-time dynamics are

$$h'(t) = Ah(t) + Bx(t), \quad y(t) = Ch(t), \quad (1)$$

where $A \in \mathbb{R}^{N \times N}$, $B \in \mathbb{R}^{N \times 1}$, and $C \in \mathbb{R}^{1 \times N}$ are learnable parameters.

With Zero-Order Hold (ZOH) discretization at step size Δ , the matrices become

$$A_d = \exp(\Delta A), \quad B_d = A^{-1}(\exp(\Delta A) - I)B. \quad (2)$$

The recurrence then updates hidden states as

$$h_t = A_d h_{t-1} + B_d x_t, \quad y_t = C h_t. \quad (3)$$

This sequential update can be parallelized by reparameterizing as a 1D convolution:

$$y = x \star \mathcal{K}, \quad \mathcal{K} = (CB, CA_d B, CA_d^2 B, \dots, CA_d^{L-1} B) \quad (4)$$

Figure 1, shows PRISM-UNet which integrates Mamba (SSMs) with convolutional neural networks.

3.2.2. Architecture

Given a 3D medical volume $\mathbf{I}_0 \in \mathbb{R}^{H \times W \times D \times C_{in}}$, a $3 \times 3 \times 3$ convolution extracts shallow features $\mathbf{X}_L \in \mathbb{R}^{H \times W \times D \times C}$. The encoder has three stages $\{L_1, L_2, L_3\}$ with resolutions $\{48^3, 24^3, 12^3\}$ and feature dimensions $\{32, 64, 128\}$. Each stage combines ResNet blocks with bidirectional Mamba Blocks for global context, with multi-scale features integrated at matching resolutions. The decoder applies CBAM-enhanced [22] upsampling with skip connections, producing refined features $\mathbf{X}_D \in \mathbb{R}^{H \times W \times D \times 2C}$ and final output $\mathbf{I} \in \mathbb{R}^{H \times W \times D \times C_{out}}$.

- Multiresolution Encoder Integration:** Input $\mathbf{I}_0$ is downsampled to generate multi-scale features at $\{1, 1/2, 1/4, 1/8\}$ resolutions. Each scale undergoes $3 \times 3 \times 3$ convolution (e.g., $\mathbf{F}_{scale}^{(0)} \in \mathbb{R}^{H \times W \times D \times 16}$), then fused with Mamba outputs via concatenation and $1 \times 1 \times 1$ channel adaptation at encoder stages $i \in \{1, 2, 3\}$.
- Bidirectional Mamba Block:** To overcome SSM directional bias, features $\mathbf{F}_0 \in \mathbb{R}^{B \times C \times H \times W \times D}$ are reshaped into spatial sequences of length $H \times W \times D$. Forward and backward branches apply LayerNorm and Mamba layers to original and reversed sequences, with outputs averaged and reshaped to $[B, C, H, W, D]$ for bidirectional context.

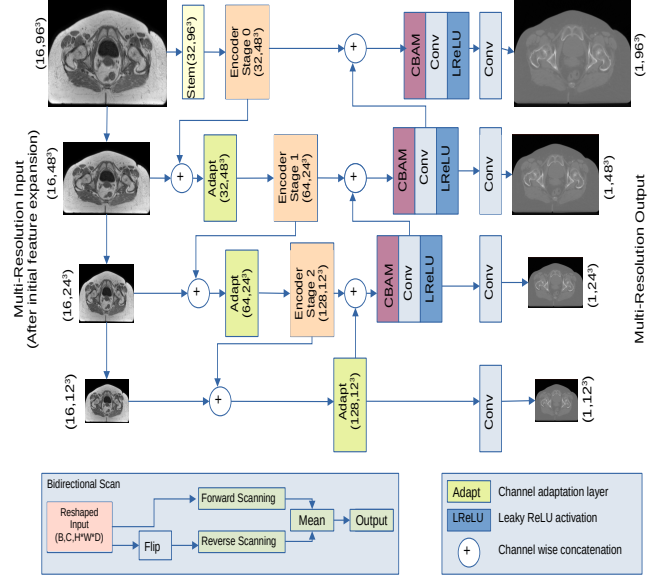


Fig. 1. The overall architecture of our PRISM-UNet.

- CBAM-Enhanced Decoder:** The decoder integrates CBAM [22] with channel and spatial attention at reduction ratios $\{8, 4, 2\}$ across stages. Spectral normalization[23] on decoder convolutions constrains the Lipschitz constant, stabilizing training.
- Discriminator:** Following previous works [6], we use a lightweight UNet architecture as a discriminator with multiresolution input processing and deep supervision enabling detailed feature-level adversarial feedback.

3.2.3. Loss formulation

Two losses are defined: generator loss (L_G) and discriminator Loss (L_D).

- Discriminator Loss:** We adopt the voxel-wise relativistic loss[24], defined as:

$$L_D = -\log(\sigma(D(I_{GT}) - \mathbb{E}_{I_{syn} \sim p_G(I)}[D(I_{syn})])) - \log(1 - \sigma(D(I_{syn}) - \mathbb{E}_{I_{GT} \sim p_{data}(I)}[D(I_{GT})])) \quad (5)$$

where $\sigma(\cdot)$ is the sigmoid function, $p_G(I)$ is the generator distribution, and $p_{data}(I)$ is the true data distribution.

- Generator Loss:** It consists of the following loss functions: voxel loss (L_{voxel}), perception loss ($L_{perception}$), ssim loss (L_{ssim}), adversarial loss (L_{adv}) and physics loss (L_{phy}) and is defined as:

$$L_G = \lambda_1 L_{voxel} + \lambda_2 L_{perception} + \lambda_3 L_{ssim} + \lambda_4 L_{adv} + \lambda_5 L_{phy} \quad (6)$$

we set the weights as $\lambda_1 = \lambda_2 = \lambda_3 = \lambda_5 = 1$ and $\lambda_4 = 0.001$ after empirical validation.

Below, we describe each component of the generator loss in detail.

- Voxel Loss:** This is the L1 loss between voxels of the real image and the synthetically generated image, defined as:

$$L_{voxel} = \|\hat{I} - I\|_1 \quad (7)$$

where $\hat{I}$ is the generated image and I is the ground truth image.

- (ii) **Perceptual Loss:** We employ a 2.5D perceptual loss using ResNet50 pretrained on RadImageNet[25]. Features from layers 4–6 are extracted to minimize differences between ground truth and generated images across axial, coronal, and sagittal views ($\mathcal{V} = \{a, c, s\}$). The loss is defined as:

$$L_{\text{perception}} = \sum_{v \in \mathcal{V}} \left\| \text{Res}(\hat{I}^{(v)}) - \text{Res}(I^{(v)}) \right\|_1 \quad (8)$$

where $\hat{I}^{(v)}$ and $I^{(v)}$ denote the generated and ground truth images in view v , and $\text{Res}(\cdot)$ extracts ResNet features.

- (iii) **SSIM Loss:** Structural similarity index defined as $\text{SSIM}(\hat{I}, I)$, where the SSIM loss is:

$$\begin{aligned} L_{\text{SSIM}} &= 1 - \text{SSIM}(\hat{I}, I) \\ &= 1 - \frac{(2\mu_{\hat{I}}\mu_I + c_1)(2\sigma_{\hat{I}I} + c_2)}{(\mu_{\hat{I}}^2 + \mu_I^2 + c_1)(\sigma_{\hat{I}}^2 + \sigma_I^2 + c_2)}. \end{aligned} \quad (9)$$

where μ and σ denote mean and standard deviation, $\sigma_{\hat{I}I}$ is the covariance, c_1 and c_2 are constants.

- (iv) **Adversarial loss:** Here we leverage the representations learned by the discriminator through an adversarial training objective.

$$\begin{aligned} L_{\text{adv}} &= -\log(\sigma(D(I_{\text{syn}}) - \mathbb{E}_{I_{\text{GT}} \sim p_{\text{data}}(I)}[D(I_{\text{GT}})])) \\ &\quad - \log(1 - \sigma(D(I_{\text{GT}}) - \mathbb{E}_{I_{\text{syn}} \sim p_G(I)}[D(I_{\text{syn}})])). \end{aligned} \quad (10)$$

where the notation follows the same definitions as in Eq. (5).

- (v) **Physics Loss based on Forward Operators:** Medical image acquisition can be modeled through forward operators that map the underlying anatomy to measured signals. For MRI, the Fourier Transform (FFT) represents k-space acquisition, where spatial frequencies are sampled according to pulse sequence trajectories ensuring synthesis respects frequency-domain characteristics of real acquisitions. For CT, the Radon Transform models X-ray projection geometry, where line integrals through tissue generate sinograms that are reconstructed via filtered backprojection enforcing consistency with the physical attenuation process.

By incorporating these forward operators into our loss formulation, we constrain the generator to produce images that are physically plausible under their respective acquisition physics.

For MRI: K-space Physics Loss — Given predicted and ground truth patches $\hat{I}, I \in \mathbb{R}^{H \times W \times D}$, the loss has two components:

K-space Domain Loss: Transform images to k-space via 3D FFT and compute L1 loss on magnitude spectra:

$$\mathcal{K}(\hat{I}) = \text{FFTShift}(\text{FFT}_3(\hat{I})) \quad (11)$$

$$\mathcal{L}_{\text{k-space}} = \left\| |\mathcal{K}(\hat{I})| - |\mathcal{K}(I)| \right\|_1 \quad (12)$$

Cyclic Consistency Loss: Simulate undersampled acquisition to enforce acquisition physics:

$$\mathcal{K}_{\text{under}} = \mathcal{K}(\hat{I}) \odot M \quad (13)$$

$$\hat{I}_{\text{recon}} = |\text{IFFT}_3(\text{IFFTShift}(\mathcal{K}_{\text{under}}))| \quad (14)$$

$$\mathcal{L}_{\text{cyclic}} = \left\| \hat{I}_{\text{recon}} - \hat{I} \right\|_1 \quad (15)$$

where $M \in \{0, 1\}^{H \times W \times D}$ is a 50% undersampling mask preserving low frequencies. The total physics loss is:

$$L_{\text{phy}} = \alpha \mathcal{L}_{\text{k-space}} + \beta \mathcal{L}_{\text{cyclic}}, \quad \alpha = 0.001, \beta = 1.0. \quad (16)$$

For CT: Radon Physics Loss — Given predicted and ground truth patches $\hat{I}, I \in \mathbb{R}^{H \times W \times D}$, the loss has two components:

Projection Domain Loss: Compute sinograms via Radon transform and measure L1 loss:

$$\mathcal{R}_{\theta}(\hat{I}) = \int_{-\infty}^{\infty} \hat{I}(t \cos \theta - s \sin \theta, t \sin \theta + s \cos \theta) dt \quad (17)$$

$$\mathcal{L}_{\text{projection}} = \left\| \mathcal{R}_{\theta}(\hat{I}) - \mathcal{R}_{\theta}(I) \right\|_1 \quad (18)$$

where $\theta \in [0, 180^\circ)$ denotes projection angles and s is the detector position.

Cyclic Consistency Loss: Enforce reconstruction physics using filtered backprojection:

$$\hat{I}_{\text{FBP}} = \mathcal{R}^{-1}(\mathcal{F}^{-1}(\mathcal{F}(\mathcal{R}_{\theta}(\hat{I})) \cdot H(\omega))) \quad (19)$$

$$\mathcal{L}_{\text{cyclic}} = \left\| \hat{I}_{\text{FBP}} - \hat{I} \right\|_1 \quad (20)$$

where $\mathcal{R}^{-1}$ is backprojection, $\mathcal{F}$ the Fourier transform, and $H(\omega)$ the ramp filter. The total physics loss is:

$$L_{\text{phy}} = \mathcal{L}_{\text{projection}} + \mathcal{L}_{\text{cyclic}}. \quad (21)$$

4. EXPERIMENTS AND ANALYSIS

4.1. Dataset and Training Setup

BraTS2021[26] with 1,251 multi-modal brain MRI scans and tumor annotations, used for T1ce→T2 and T2→FLAIR translation; and **SynthRAD2023**[27], from which 180 pelvis MRI→CT and 180 CBCT→CT samples are selected. The dataset is split into 75% training and 25% testing set as in previous works. Training employs 96^3 patches, batch size 3, FP16 precision, Adam optimizer ($\text{lr}=1 \times 10^{-4}$) with MultiStepLR scheduler. Each task is run five times and average result is taken. Experiments run on a single H200 GPU, requiring ~36 GB VRAM (2.1 GB for inference). With smaller batch size, training fits in 12.8 GB, allowing use on 16 GB GPUs.

4.2. Results and Comparisons

PRISM-UNet was evaluated across four medical image translation tasks covering brain and pelvis anatomies. Table 1 presents quantitative comparisons using SSIM, PSNR, NMSE, and Dice metrics, while Figure 2 shows qualitative results. Our method consistently outperforms all baselines across tasks.

For BraTS2021 T1ce→T2, PRISM-UNet achieves SSIM of 0.953 and PSNR of 29.305, representing improvements over 3D-mDAUNet (0.937, 27.897). The progressive PSNR improvement from ALDM (25.880) → ResViT (26.190) → Ea-GAN (27.376) → 3D-mDAUNet (27.897) → Ours (29.305) demonstrates genuine methodological advances. For SynthRAD pelvis tasks, our method shows substantial NMSE reductions: MRI→CT (0.070 vs. 0.073 for 3D-mDAUNet) and CBCT→CT (0.045 vs. 0.048). Qualitatively, Figure 2 reveals that our outputs exhibit sharper anatomical boundaries, better tissue contrast preservation, and reduced blurring compared to baselines. In MRI→CT translation, bone boundaries

are more clearly delineated. For CBCT→CT, our approach effectively removes cone-beam artifacts while maintaining structural integrity.

For downstream BraTS2021 tumor segmentation using synthetic FLAIR, we achieve Dice 0.884 vs. 3D-mDAUNet (0.878) and EaGAN (0.866), demonstrating effective real-world generalization.

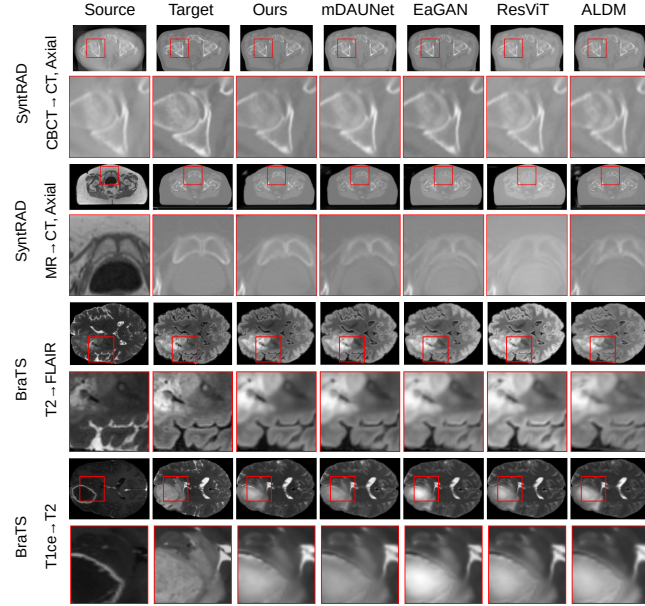


Fig. 2. Qualitative comparison across four translation tasks showing source, target, and synthesized outputs from all methods. Our approach produces sharper anatomical boundaries and superior tissue contrast compared to baselines.

Table 1. Quantitative results on four tasks. Best in **bold**, second best underlined. Dice score omitted for SynthRAD (no ground truth).

Experiment	Method	SSIM ↑	PSNR ↑	NMSE ↓	DICE ↑
BraTS2021 Brain T1ce→T2	ResViT	0.907±0.90	26.190±2.85	0.210±0.10	0.721±0.15
	ALDM	0.902±0.05	25.880±2.10	0.201±0.10	0.719±0.15
	Ea-GAN	0.922±0.05	27.376±2.46	0.193±0.12	0.732±0.17
	3D-mDAUNet	0.937±0.06	27.897±2.61	0.189±0.30	0.750±0.19
	Ours	0.953±0.02	29.305±2.50	0.133±0.21	0.768±0.18
BraTS2021 Brain T2→FLAIR	ResViT	0.887±0.05	27.825±2.18	0.162±0.19	0.849±0.13
	ALDM	0.891±0.04	28.212±3.10	0.135±0.15	0.819±0.14
	Ea-GAN	<u>0.942±0.05</u>	29.972±3.34	0.109±0.12	0.866±0.12
	3D-mDAUNet	0.938±0.03	29.549±3.11	0.112±0.13	0.878±0.11
	Ours	0.945±0.05	29.950±3.22	0.110±0.13	0.884±0.11
SynthRAD Pelvis T1 MRI→CT	ResViT	0.844±0.05	28.066±1.85	0.112±0.05	-
	ALDM	0.824±0.06	28.395±1.88	0.108±0.04	-
	Ea-GAN	0.853±0.06	27.492±2.55	0.131±0.03	-
	3D-mDAUNet	<u>0.888±0.03</u>	<u>29.787±1.79</u>	<u>0.073±0.03</u>	-
	Ours	0.898±0.03	30.010±1.34	0.070±0.03	-
SynthRAD Pelvis CBCT→CT	ResViT	0.894±0.07	30.425±1.35	0.074±0.03	-
	ALDM	0.888±0.05	31.105±3.05	0.073±0.04	-
	Ea-GAN	0.832±0.04	25.466±2.88	0.210±0.07	-
	3D-mDAUNet	0.914±0.04	31.995±3.22	0.048±0.03	-
	Ours	0.926±0.04	32.225±3.33	0.045±0.03	-

An ablation study is done using a leave-one-out strategy. For this analysis, we focus on the MRI-to-CT translation task using the SynthRAD2023 dataset. The results are summarized in Table 2.

Table 2. Analysis of the contribution of each component proposed in our study.

Modules				Image Quality Assessment		
Bi-Scan	Physics Loss	UNet D.	Relativistic	SSIM ↑	PSNR ↑	NMSE ↓
✓				0.850±0.02	27.5±1.10	0.105±0.02
✓		✓		0.880±0.04	28.8±1.25	0.090±0.03
✓	✓	✓		0.890±0.03	29.2±1.18	0.085±0.03
✓	✓	✓	✓	0.898±0.03	30.010±1.34	0.070±0.03

5. LIMITATIONS AND FUTURE WORK

While PRISM-UNet demonstrates strong performance, several limitations warrant consideration. First, patch-wise processing may introduce subtle boundary artifacts at patch interfaces despite Gaussian blending. Second, registered and paired training data is assumed, which may not always be available. Third, physics losses assume idealized acquisition models (perfect k-space sampling for MRI, parallel-beam geometry for CT) that may not fully capture real-world scanner variations and noise characteristics.

Future work will address full-volume processing through memory-efficient architectures, extend to unpaired learning scenarios via cycle-consistency approaches, incorporate scanner-specific physics models, and conduct multi-site clinical validation across broader anatomical regions.

6. CONCLUSION

In this work, we proposed PRISM-UNet, a novel medical image translation framework that integrates Mamba-based state-space modeling with multiresolution guidance and physics-based constraints. Extensive evaluations demonstrate consistent state-of-the-art performance while preserving structural fidelity. The framework exhibits strong synthetic-to-real generalization in downstream segmentation, confirming its clinical applicability. Overall, PRISM-UNet establishes a robust foundation for scalable medical image synthesis, enabling improved automated diagnosis, precision treatment planning, and integration into clinical workflows.

7. COMPLIANCE WITH ETHICAL STANDARDS

This research study was conducted retrospectively using human subject data made available in open access by BraTS2021[26] and SynthRAD2023[27]. Ethical approval was not required as confirmed by the license attached with the open access data.

8. REFERENCES

- [1] Xiang Li, Yuchen Jiang, Juan J Rodriguez-Andina, Hao Luo, Shen Yin, and Okyay Kaynak, “When medical images meet generative adversarial network: recent development and research opportunities,” *Discover Artificial Intelligence*, vol. 1, pp. 1–20, 2021.
- [2] Jelmer M Wolterink, Anna M Dinkla, Mark HF Savenije, Peter R Seevinck, Cornelis AT van den Berg, and Ivana Išgum, “Deep mr to ct synthesis using unpaired data,” in *Simulation*

and Synthesis in Medical Imaging: Second International Workshop, SASHIMI 2017, Held in Conjunction with MICCAI 2017, Québec City, QC, Canada, September 10, 2017, Proceedings 2. Springer, 2017, pp. 14–23.

- [3] Jun-Yan Zhu, Taesung Park, Phillip Isola, and Alexei A Efros, “Unpaired image-to-image translation using cycle-consistent adversarial networks,” in *Proceedings of the IEEE international conference on computer vision*, 2017, pp. 2223–2232.
- [4] Denis Prokopenko, Joël Valentin Stadelmann, Heinrich Schulz, Steffen Renisch, and Dmitry V Dylov, “Unpaired synthetic image generation in radiology using gans,” in *Artificial Intelligence in Radiation Therapy: First International Workshop, AIRT 2019, Held in Conjunction with MICCAI 2019, Shenzhen, China, October 17, 2019, Proceedings 1*. Springer, 2019, pp. 94–101.
- [5] Alaa Abu-Srhan, Israa Almallahi, Mohammad AM Abushariah, Waleed Mahafza, and Omar S Al-Kadi, “Paired-unpaired unsupervised attention guided gan with transfer learning for bidirectional brain mr-ct synthesis,” *Computers in Biology and Medicine*, vol. 136, pp. 104763, 2021.
- [6] Juhung Ha, Jong Sung Park, David Crandall, Eleftherios Garyfallidis, and Xuhong Zhang, “Multi-resolution guided 3d gans for medical image translation,” in *2025 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*. IEEE, 2025, pp. 4342–4351.
- [7] Onat Dalmaz, Mahmut Yurt, and Tolga Çukur, “Resvit: residual vision transformers for multimodal medical image synthesis,” *IEEE Transactions on Medical Imaging*, vol. 41, no. 10, pp. 2598–2614, 2022.
- [8] Biting Yu, Luping Zhou, Lei Wang, Yinghuan Shi, Jurgen Fripp, and Pierrick Bourgeat, “Ea-gans: edge-aware generative adversarial networks for cross-modality mr image synthesis,” *IEEE transactions on medical imaging*, vol. 38, no. 7, pp. 1750–1762, 2019.
- [9] Jonghun Kim and Hyunjin Park, “Adaptive latent diffusion model for 3d medical image to image translation: Multi-modal magnetic resonance imaging study,” in *Proceedings of the IEEE/CVF Winter conference on applications of computer vision*, 2024, pp. 7604–7613.
- [10] Abbas Bahrami, Alireza Karimian, Emad Fatemizadeh, Hossein Arabi, and Habib Zaidi, “A novel convolutional neural network with high convergence rate: Application to ct synthesis from mr images,” in *2019 IEEE Nuclear Science Symposium and Medical Imaging Conference (NSS/MIC)*. IEEE, 2019, pp. 1–3.
- [11] Abass Bahrami, Alireza Karimian, Emad Fatemizadeh, Hossein Arabi, and Habib Zaidi, “A new deep convolutional neural network design with efficient learning capability: Application to ct image synthesis from mri,” *Medical physics*, vol. 47, no. 10, pp. 5158–5171, 2020.
- [12] Karim Armanious, Chenming Jiang, Marc Fischer, Thomas Küstner, Tobias Hepp, Konstantin Nikolaou, Sergios Gatidis, and Bin Yang, “Medgan: Medical image translation using gans,” *Computerized medical imaging and graphics*, vol. 79, pp. 101684, 2020.
- [13] Karim Armanious, Chenming Jiang, Sherif Abdulatif, Thomas Küstner, Sergios Gatidis, and Bin Yang, “Unsupervised medical image translation using cycle-medgan,” in *2019 27th European signal processing conference (EUSIPCO)*. IEEE, 2019, pp. 1–5.
- [14] Bo Zhao, Tingting Cheng, Xueren Zhang, Jingjing Wang, Hong Zhu, Rongchang Zhao, Dengwang Li, Zijian Zhang, and Gang Yu, “Ct synthesis from mr in the pelvic area using residual transformer conditional gan,” *Computerized medical imaging and graphics*, vol. 103, pp. 102150, 2023.
- [15] Sheng Zhao, Changran Geng, Chang Guo, Feng Tian, and Xiaobin Tang, “Saru: a self-attention resunet to generate synthetic ct images for mr-only bnct treatment planning,” *Medical Physics*, vol. 50, no. 1, pp. 117–127, 2023.
- [16] Pu Zhao, Hong Pan, and Siyu Xia, “Mri-trans-gan: 3d mri cross-modality translation,” in *2021 40th Chinese Control Conference (CCC)*. IEEE, 2021, pp. 7229–7234.
- [17] Zolnamar Dorjsembe, Hsing-Kuo Pao, Sodtavilan Odonchimed, and Furen Xiao, “Conditional diffusion models for semantic 3d brain mri synthesis,” *IEEE Journal of Biomedical and Health Informatics*, vol. 28, no. 7, pp. 4084–4093, 2024.
- [18] Jiacheng Ruan, Jincheng Li, and Suncheng Xiang, “Vm-unet: Vision mamba unet for medical image segmentation,” *arXiv preprint arXiv:2402.02491*, 2024.
- [19] Jun Ma, Feifei Li, and Bo Wang, “U-mamba: Enhancing long-range dependency for biomedical image segmentation,” *arXiv preprint arXiv:2401.04722*, 2024.
- [20] Albert Gu, Karan Goel, and Christopher Ré, “Efficiently modeling long sequences with structured state spaces,” *arXiv preprint arXiv:2111.00396*, 2021.
- [21] Albert Gu and Tri Dao, “Mamba: Linear-time sequence modeling with selective state spaces,” *arXiv preprint arXiv:2312.00752*, 2023.
- [22] Sanghyun Woo, Jongchan Park, Joon-Young Lee, and In So Kweon, “Cbam: Convolutional block attention module,” in *Proceedings of the European conference on computer vision (ECCV)*, 2018, pp. 3–19.
- [23] Takeru Miyato, Toshiki Kataoka, Masanori Koyama, and Yuichi Yoshida, “Spectral normalization for generative adversarial networks,” *arXiv preprint arXiv:1802.05957*, 2018.
- [24] Alexia Jolicoeur-Martineau, “The relativistic discriminator: a key element missing from standard gan,” *arXiv preprint arXiv:1807.00734*, 2018.
- [25] Xueyan Mei, Zelong Liu, Philip M Robson, Brett Marinelli, Mingqian Huang, Amish Doshi, Adam Jacobi, Chendi Cao, Katherine E Link, Thomas Yang, et al., “Radimagenet: an open radiologic deep learning research dataset for effective transfer learning,” *Radiology: Artificial Intelligence*, vol. 4, no. 5, pp. e210315, 2022.
- [26] Ujjwal Baid, Satyam Ghodasara, Suyash Mohan, Michel Bilello, Evan Calabrese, Errol Colak, Keyvan Farahani, Jayashree Kalpathy-Cramer, Felipe C Kitamura, Sarthak Pati, et al., “The rsna-asnr-miccai brats 2021 benchmark on brain tumor segmentation and radiogenomic classification,” *arXiv preprint arXiv:2107.02314*, 2021.
- [27] Adrian Thummerer, Erik van der Bijl, Arthur Galapon Jr, Joost JC Verhoeff, Johannes A Langendijk, Stefan Both, Cornelis (Nico) AT van den Berg, and Matteo Maspero, “Synthrad2023 grand challenge dataset: Generating synthetic ct for radiotherapy,” *Medical physics*, vol. 50, no. 7, pp. 4664–4674, 2023.